Data Mining



Data Reduction

Rivka Gadot

Partly based on Introduction to Data Mining 2nd edition and Data Mining concepts and techniques

Data Reduction

- **Data reduction**:
  - Obtain a reduced representation of the data set that is much smaller in volume but yet produces the same (or almost the same) analytical results
- **Why data reduction?**
  - A database/data warehouse may store terabytes of data. Complex data analysis may take a very long time to run on the complete data set.
- **Data reduction strategies**
  - Dimensionality reduction, e.g., remove unimportant attributes
  - Numerosity reduction (some simply call it: Data Reduction)
  - Data compression

Dimensionality reduction methods

transform the original data onto a smaller space:

- Wavelet Transforms
- Principal Components Analysis (PCA)
- t-SNE
- Autoencoders

removed irrelevant, weakly relevant, or redundant attributes:

- Attribute subset selection

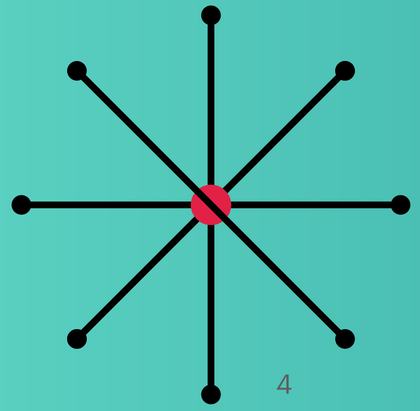
Principal Components Analysis (PCA)

- PCA is an unsupervised statistical method used to decompose multidimensional data.
- Its primary objective is to reduce a multidimensional dataset into a set of arbitrary variables by selecting essential features from the original dataset.
- "Bellman's Curse of Dimensionality" is a concept in ML that arises when dealing with high-dimensional data.

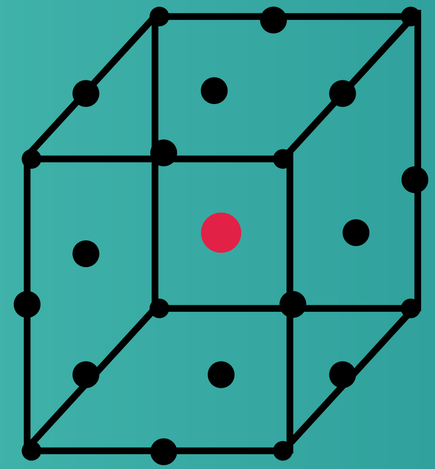
1-d



2-d



3-d



Principal Components Analysis (PCA)

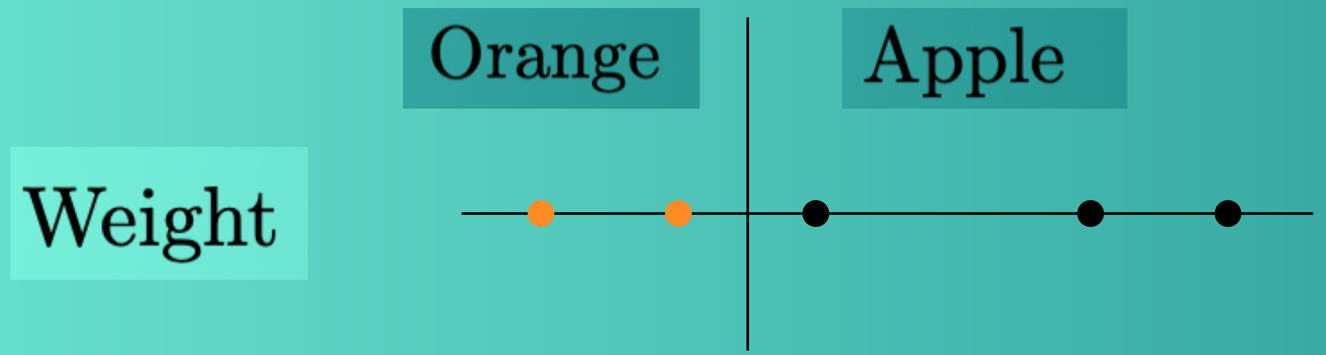
Example

Weight	Color Score	Diameter	Sugar Content	Fruit Type
150	0.75	7.3	0.9	Apple
180	0.65	8.0	0.8	Apple
130	0.80	7.0	0.7	Orange
170	0.70	7.5	0.85	Apple
140	0.85	6.5	0.75	Orange

Principal Components Analysis (PCA)

Example

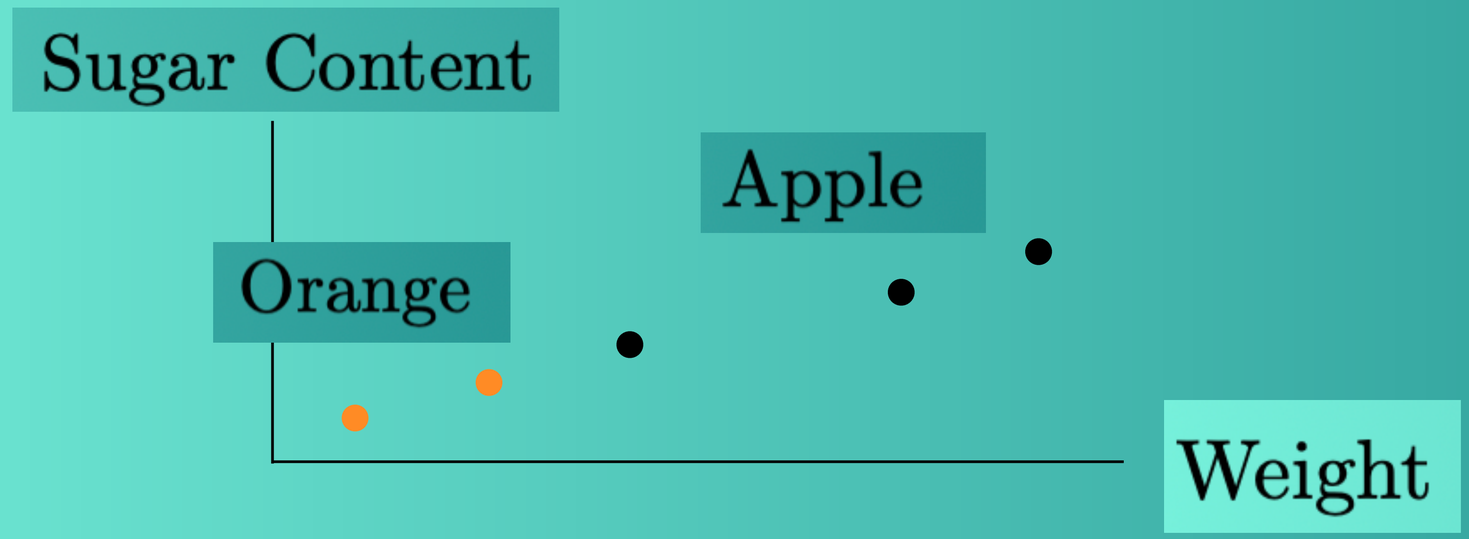
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Principal Components Analysis (PCA)

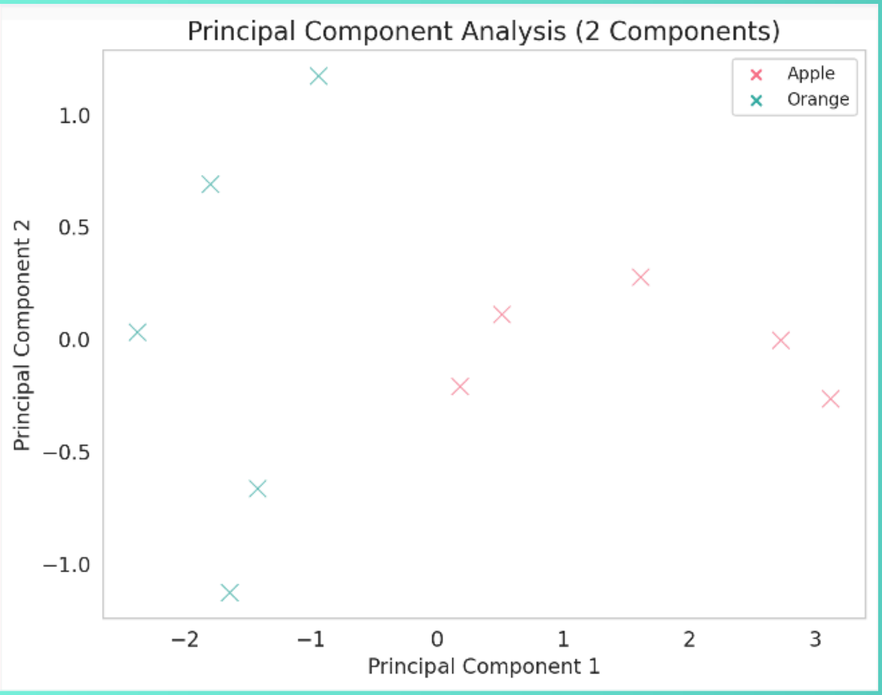
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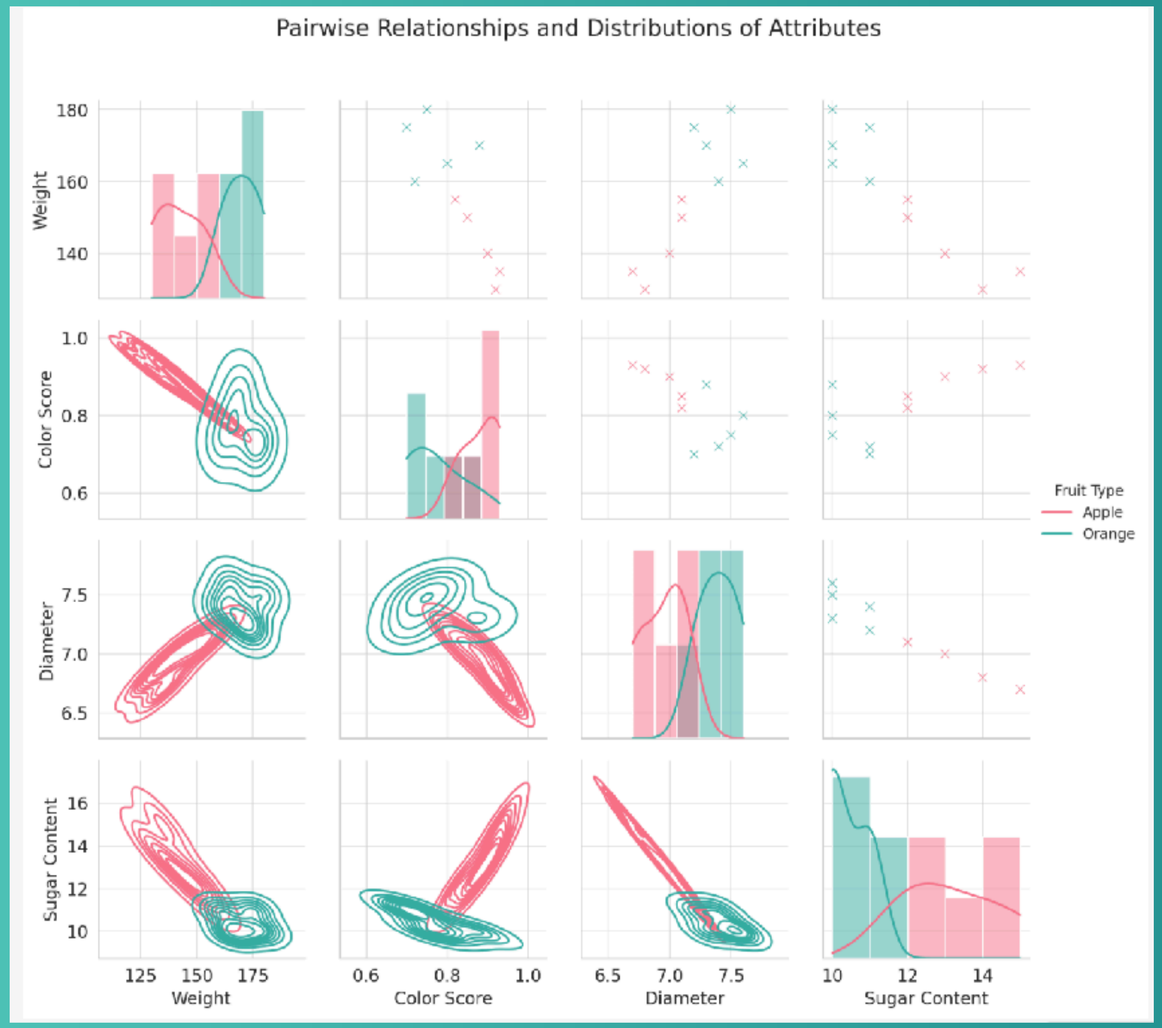
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Principal Components Analysis (PCA)

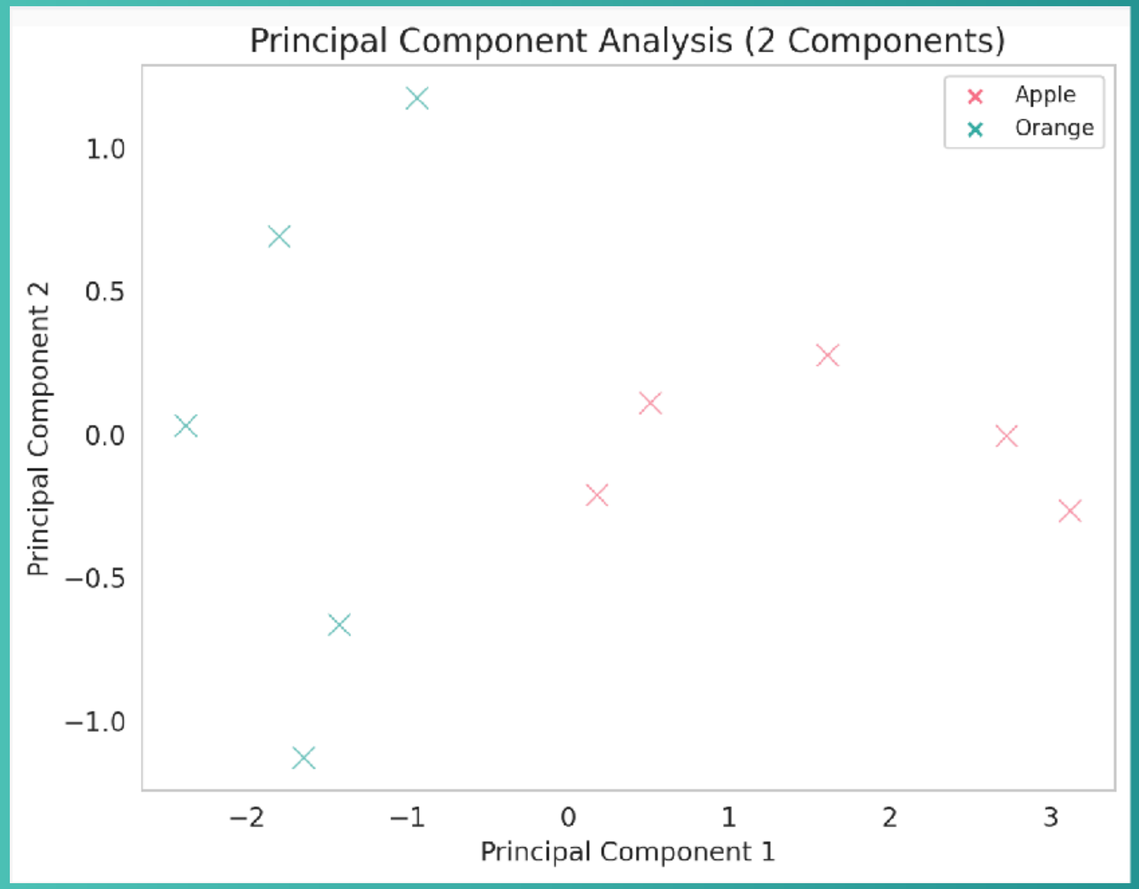
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Principal Components Analysis (PCA)

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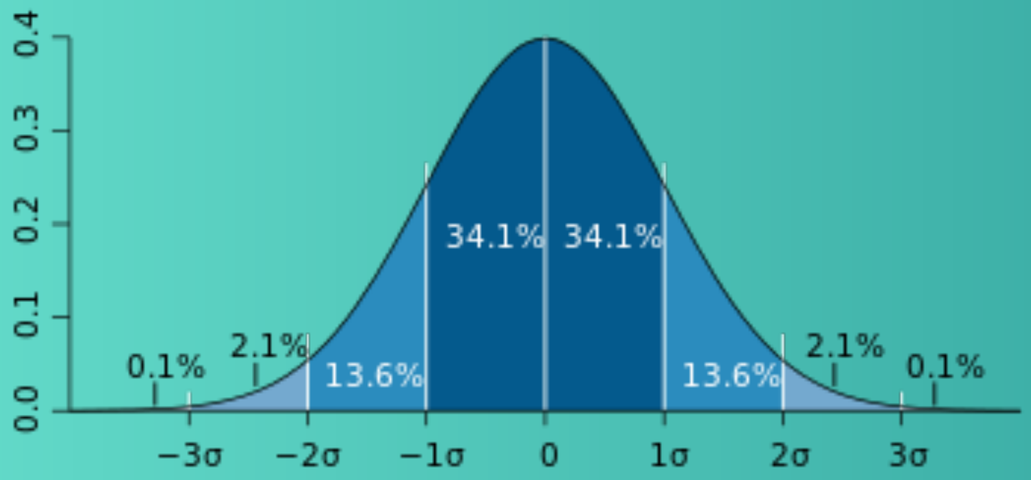


- Principal Component 1 explains approximately 86.45% of the variance.
- Principal Component 2 explains 9.42% of the variance.

By projecting our 4-dimensional data down to a 2-dimensional space, we've retained 95.87% of the variability of the original data

Principal Components Analysis (PCA)

- PCA is deterministic (not stochastic).



Cons:

- PCA is a linear algorithm
- Sensitive to outliers

Principal Components Analysis (PCA)

- PCA can be employed for reducing the dimensionality of a dataset while preserving the fundamental relationships between data points.
- Despite its flexibility, speed, and interpretability, PCA may not yield optimal results when dealing with data containing nonlinear relationships.

Principal Components Analysis (PCA)

Steps:

- **Standardize the Data**.
- **Compute the Covariance Matrix**.
- **Find Eigenvectors and Eigenvalues**.
- **Sort and Select Principal Components**: Choose the top k principal components based on the eigenvalues.
- **Project Data onto Lower-Dimensional Subspace**: Transform the original data into the new subspace created by the principal components.

Principal Components Analysis (PCA)

Example

Weight	Color Score	Diameter	Sugar Content	Fruit Type
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Principal Components Analysis (PCA)

step 1: Standardize the Data

Weight
150
180
130
170
140

$$\mu = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{150 + 180 + 130 + 170 + 140 + \dots}{n}$$

$$\sigma = \sqrt{\frac{\sum_{i=1}^{n} (x_i - \mu)^2}{n}}$$

$$\mu = 156.0, \sigma = 16.25$$

$$z_i = \frac{x_i - \mu}{\sigma}$$

Principal Components Analysis (PCA)

step 1: Standardize the Data

Weight
150
180
130
170
140

$$z_1 = \frac{150 - \mu}{\sigma} = \frac{150 - 156.0}{16.25} = -0.369$$

$$z_2 = \frac{180 - \mu}{\sigma} = \frac{180 - 156.0}{16.25} = 1.477$$

$$z_3 = \frac{130 - \mu}{\sigma} = \frac{130 - 156.0}{16.25} = -1.600$$

$$z_4 = \frac{170 - \mu}{\sigma} = \frac{170 - 156.0}{16.25} = 0.862$$

$$z_5 = \frac{140 - \mu}{\sigma} = \frac{140 - 156.0}{16.25} = -0.985$$

And so on with the other attributes...

Principal Components Analysis (PCA)

step 2: Compute the Covariance Matrix

x_i - is the $i-th$ observation (a row vector of length d)

$\overline{x}$ - is the mean vector of the dataset (also of length d)

n - is the total number of observations

T - denotes the transpose of a matrix

$$\mathbf{S} = \frac{1}{n}\sum_{i=1}^{n}(\mathbf{x}_i - \overline{\mathbf{x}})(\mathbf{x}_i - \overline{\mathbf{x}})^T$$

Principal Components Analysis (PCA)

step 2: Compute the Covariance Matrix

Initialize an empty covariance matrix: $\mathbf{S} = 0_{d \times d}$

Calculate the mean vector:

$$\overline{\mathbf{x}} = \frac{1}{n} \sum_{i=1}^{n} \mathbf{x}_i$$

Compute the covariance matrix:

$$\mathbf{S} = \frac{1}{n} \sum_{i=1}^{n} (\mathbf{x}_i - \overline{\mathbf{x}})(\mathbf{x}_i - \overline{\mathbf{x}})^T$$

Principal Components Analysis (PCA)

step 2: Compute the Covariance Matrix

$$\begin{bmatrix} (x_{i1} - \bar{x}_1) \\ (x_{i2} - \bar{x}_2) \\ \vdots \\ (x_{id} - \bar{x}_d) \end{bmatrix} \begin{bmatrix} (x_{i1} - \bar{x}_1) & (x_{i2} - \bar{x}_2) & \dots & (x_{id} - \bar{x}_d) \end{bmatrix}$$

$$\begin{bmatrix} (x_{i1} - \bar{x}_1)(x_{i1} - \bar{x}_1) & (x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2) & \dots & (x_{i1} - \bar{x}_1)(x_{id} - \bar{x}_d) \\ (x_{i2} - \bar{x}_2)(x_{i1} - \bar{x}_1) & (x_{i2} - \bar{x}_2)(x_{i2} - \bar{x}_2) & \dots & (x_{i2} - \bar{x}_2)(x_{id} - \bar{x}_d) \\ \vdots & \vdots & \ddots & \vdots \\ (x_{id} - \bar{x}_d)(x_{i1} - \bar{x}_1) & (x_{id} - \bar{x}_d)(x_{i2} - \bar{x}_2) & \dots & (x_{id} - \bar{x}_d)(x_{id} - \bar{x}_d) \end{bmatrix}$$

Principal Components Analysis (PCA)

step 2: Compute the Covariance Matrix

The solution is:

$$\begin{bmatrix} 1 & -0.91499142 & 0.85766302 & 0.533745 \\ -0.91499142 & 1 & -0.98915848 & -0.5 \\ 0.85766302 & -0.98915848 & 1 & 0.45218674 \\ 0.533745 & -0.5 & 0.45218674 & 1 \end{bmatrix}$$

We will solve it in the lab

Principal Components Analysis (PCA)

step 3: Find Eigenvectors and Eigenvalues

Given a square matrix **A**, a non-zero vector **v**:

v is an eigenvector of **A** if it satisfies the following equation:

$$Av = \lambda v$$

λ is a scalar called an eigenvalue corresponding to the eigenvector **v**

Principal Components Analysis (PCA)

step 3: Find Eigenvectors and Eigenvalues

Steps to Calculate Eigenvalues λ and Eigenvectors $\mathbf{v}$:

Find the Eigenvalues:

$$det(A - \lambda I) = 0$$

Where:

$\mathbf{A}$ is the matrix (e.g., the covariance matrix),

λ is a scalar (the eigenvalue),

$\mathbf{I}$ is the identity matrix of the same shape as $\mathbf{A}$,

det represents the determinant of a matrix.

Principal Components Analysis (PCA)

step 3: Find Eigenvectors and Eigenvalues

Steps to Calculate Eigenvalues λ and Eigenvectors $\mathbf{v}$:

Find the Eigenvectors:

find the eigenvectors by substituting each eigenvalue λ back into the equation: $(A - \lambda I)v = 0$ And solving for $\mathbf{v}$.

Note that this equation usually has infinite solutions, and we typically normalize $\mathbf{v}$ to have unit length.

Principal Components Analysis (PCA)

step 3: Find Eigenvectors and Eigenvalues

Example Calculation:

$$\mathbf{A} = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$$

Compute Eigenvalues:

Find the characteristic equation by solving $\det(\mathbf{A} - \lambda\mathbf{I}) = 0$

where $\mathbf{I}$ is the identity matrix, and λ represents the eigenvalues.

Principal Components Analysis (PCA)

step 3: Find Eigenvectors and Eigenvalues

$$\det\left(\begin{bmatrix}4 & 2\\ 2 & 4\end{bmatrix} - \lambda\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\right) = 0$$

$$\det\left(\begin{bmatrix}4-\lambda & 2\\ 2 & 4-\lambda\end{bmatrix}\right) = 0$$

$$(4-\lambda)^2 - 4 = 0$$

Solving for λ gives the eigenvalues $\lambda_1 = 6$ and $\lambda_2 = 2$

Principal Components Analysis (PCA)

Compute Eigenvectors:

Find the eigenvectors by substituting the eigenvalues back into the equation

$(A - \lambda \mathbf{I})\mathbf{v} = 0$ and solving for $\mathbf{v}$.

For $\lambda_1 = 6$:

$$\begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} \mathbf{v}_1 - 6 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{v}_1 = 0 \longrightarrow \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} \mathbf{v}_1 = 0$$

$$-2v_{11} + 2v_{12} = 0 \longrightarrow v_{11} = v_{12}$$

If we take $v_{11} = 1$, then $v_{12} = 1 \longrightarrow \mathbf{v}_1 = [1,1]$

Principal Components Analysis (PCA)

Compute Eigenvectors:

Find the eigenvectors by substituting the eigenvalues back into the equation

$(A - \lambda \mathbf{I})\mathbf{v} = 0$ and solving for $\mathbf{v}$.

For $\lambda_2 = 2$:

$$\begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} \mathbf{v}_2 - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \mathbf{v}_2 = 0 \longrightarrow \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \mathbf{v}_2 = 0$$

$$2v_{21} + 2v_{22} = 0 \longrightarrow v_{11} = -v_{12}$$

If we take $v_{21} = 1$, then $v_{22} = -1$ $\longrightarrow$ $\mathbf{v}_2 = [1, -1]$

Principal Components Analysis (PCA)

step 4: Sort and Select Principal Components

```
Step 3: Computing Eigenvalues and Eigenvectors
Eigenvalues:
 [3.17959814e+00 6.70568521e-01 1.47169507e-01 2.66382996e-03]

Eigenvectors:
 [[-0.5297762   0.11194245 -0.82064796 -0.18260064]
 [ 0.54844555 -0.22954359 -0.21283638 -0.77538242]
 [-0.53333548  0.28673474  0.51786806 -0.60427571]
 [-0.36619383 -0.92334274  0.11423781 -0.01702893]]
```

We will solve it in the lab

Principal Components Analysis (PCA)

step 4: Sort and Select Principal Components

Sort the eigenvectors by decreasing eigenvalues and choose the first k eigenvectors, where k is the number of dimensions of the new feature space $k \leq d$ where d is the number of original dimensions

```
Step 3: Computing Eigenvalues and Eigenvectors
Eigenvalues:
 [3.17959814e+00 6.70568521e-01 1.47169507e-01 2.66382996e-03]

Eigenvectors:
 [[-0.5297762   0.11194245 -0.82064796 -0.18260064]
 [ 0.54844555 -0.22954359 -0.21283638 -0.77538242]
 [-0.53333548  0.28673474  0.51786806 -0.60427571]
 [-0.36619383 -0.92334274  0.11423781 -0.01702893]]
```

Principal Components Analysis (PCA)

step 4: Sort and Select Principal Components

```
Step 4: Selecting Principal Components
Selected Eigenvectors (Principal Components):
 [[-0.5297762   0.11194245]
 [ 0.54844555 -0.22954359]
 [-0.53333548  0.28673474]
 [-0.36619383 -0.92334274]]
```

Principal Components Analysis (PCA)

step 5: Project Data onto Lower-Dimensional Subspace

Form the Projection Matrix:

Construct a projection matrix W from the selected k eigenvectors.

The projection matrix W is formed by concatenating the top k eigenvectors in columns:

$$W = [\mathbf{e}_1, \mathbf{e}_2, \ldots, \mathbf{e}_k]$$

Principal Components Analysis (PCA)

step 5: Project Data onto Lower-Dimensional Subspace

Project the Data:

Use the projection matrix to transform the original dataset X to obtain a

k-dimensional feature subspace Y.

To project the data onto this lower-dimensional subspace, we compute the dot product of the data matrix X of size $n \times d$ and the projection matrix W of size $d \times k$:

$$Y = XW$$

X is the original data of shape (n, d) (n samples and d features),
W is the projection matrix of shape (d, k)
Y is the transformed data of shape (n, k)

Principal Components Analysis (PCA)

step 5: Project Data onto Lower-Dimensional Subspace

Example Calculation:

$$\mathbf{X} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 8 \end{bmatrix} \qquad \mathbf{W} = \begin{bmatrix} 0.707 & -0.707 \\ 0.707 & 0.707 \end{bmatrix}$$

$$y_{11} = (1)(0.707) + (2)(0.707) = 2.121$$
$$y_{12} = (1)(-0.707) + (2)(0.707) = 0.707$$
$$y_{21} = (3)(0.707) + (4)(0.707) = 4.949$$
$$y_{22} = (3)(-0.707) + (4)(0.707) = 0.707$$
$$y_{31} = (5)(0.707) + (6)(0.707) = 7.777$$
$$y_{32} = (5)(-0.707) + (6)(0.707) = 0.707$$

$$\longrightarrow \quad \mathbf{Y} = \begin{bmatrix} 2.121 & 0.707 \\ 4.949 & 0.707 \\ 7.777 & 0.707 \end{bmatrix}$$

Principal Components Analysis (PCA)

step 5: Project Data onto Lower-Dimensional Subspace

```
Step 5: Projecting Data onto Lower-Dimensional Subspace
Projected Data:
 [[-0.44625456 -1.30702551]
 [-2.30697901  0.90557583]
 [ 1.86832554  0.84965629]
 [-1.35956208 -0.25649886]
 [ 2.24447011 -0.19170776]]
```

Classification

use the resulting principal components to predict new examples

- Standardize the new example using the same StandardScaler that was fitted on the original dataset.
- Project the standardized features onto the principal components using the PCA transformation we've determined.
- Train a classifier on the PCA-reduced dataset.
- Use the trained classifiers to predict.

Classification

use the resulting principal components to predict new examples

Weight	Color Score	Diameter	Sugar Content	Fruit Type
150	0.75	7.3	0.9	Apple
180	0.65	8.0	0.8	Apple
130	0.80	7.0	0.7	Orange
170	0.70	7.5	0.85	Apple
140	0.85	6.5	0.75	Orange

The example to classify:

160	0.77	7.4	0.82

Classification

use the resulting principal components to predict new examples

1. Feature Means and Standard Deviations:

Weight: mean = 154, std = 18.547

Color Score: mean = 0.75, std = 0.071

Diameter: mean = 7.26, std = 0.5.

Sugar Content: mean = 0.8, std = 0.071

2. Standardization:

$$z_{Weight} = \frac{160 - 154}{18.547} = 0.323$$

$$z_{Color} = \frac{0.77 - 0.75}{0.071} = 0.283$$

$$z_{Diameter} = \frac{7.4 - 7.26}{0.5} = 0.280$$

$$z_{Sugar} = \frac{0.82 - 0.8}{0.071} = 0.283$$

Classification

use the resulting principal components to predict new examples

```
Step 4: Selecting Principal Components
Selected Eigenvectors (Principal Components):
 [[-0.5297762   0.11194245]
 [ 0.54844555 -0.22954359]
 [-0.53333548  0.28673474]
 [-0.36619383 -0.92334274]]
```

$$z_{Weight} = 0.323$$

$$z_{Color} = 0.283$$

$$z_{Diameter} = 0.280$$

$$z_{Sugar} = 0.283$$

Projection onto PC1:

PC1=(0.323×–0.5298)+(0.283×0.5484)+(0.280×–0.5333)+(0.283×–0.3662)=-0.2689

Projection onto PC2:

PC2=(0.323×0.1119)+(0.283×–0.2295)+(0.280×0.2867)+(0.283×–0.9233)=-0.2098

Classification

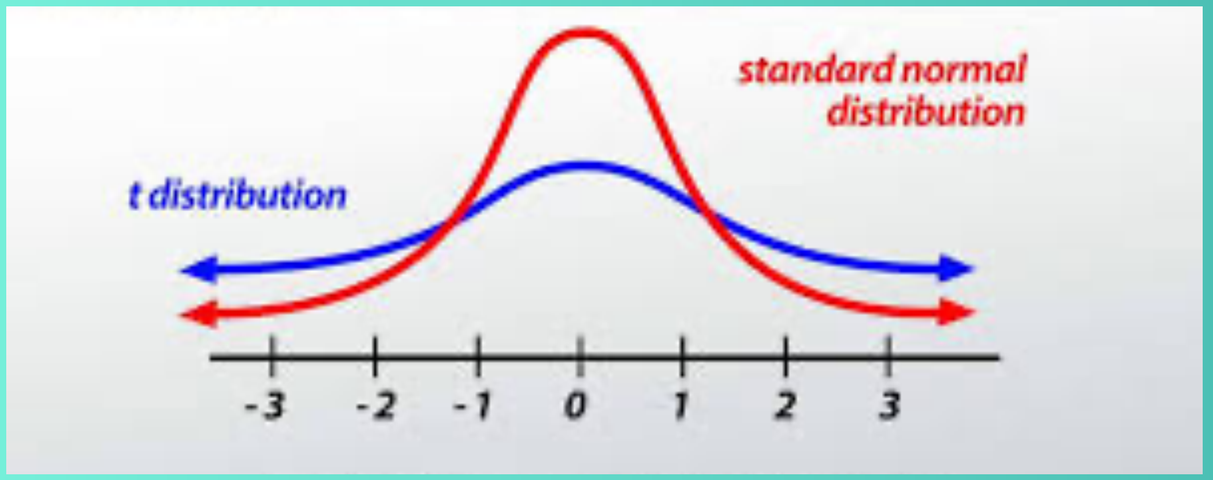
use the resulting principal components to predict new examples

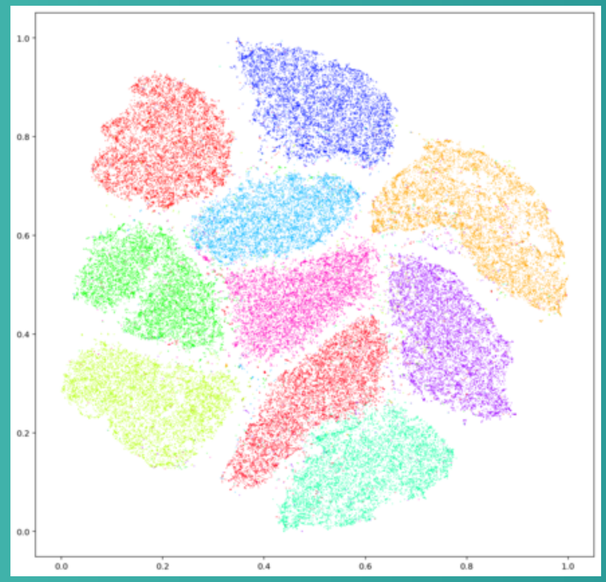
PC1=-0.2689

PC2=-0.2098

t-distributed Stochastic Neighbor Embedding (t-SNE)

- t-SNE maps the data into a lower-dimensional space where a short distance between two points signifies they were close in the original space.
- t-SNE is stochastic.
- Based on t-student and not Gaussian





t-distributed Stochastic Neighbor Embedding (t-SNE)

$$p_{j|i} = \frac{\exp(-\| x_i - x_j \|^2 / 2\sigma_i^2)}{\sum_{k \neq i} \exp(-\| x_k - x_j \|^2 / 2\sigma_i^2)}$$

The probabilities in the original space

$$q_{j|i} = \frac{\exp(-\| y_i - y_j \|^2)}{\sum_{k \neq i} \exp(-\| y_k - y_j \|^2)}$$

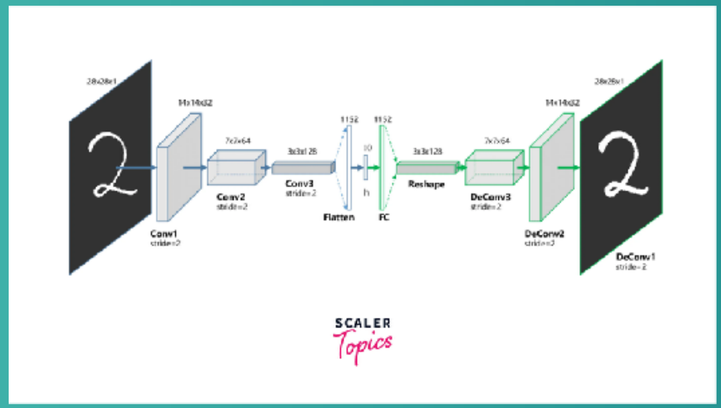
The probabilities in the space of represented points

$$C = \sum_{i,j} p_{j|i} \log \frac{p_{j|i}}{q_{j|i}}$$

the divergence between the distributions is minimized using a gradient descent over y

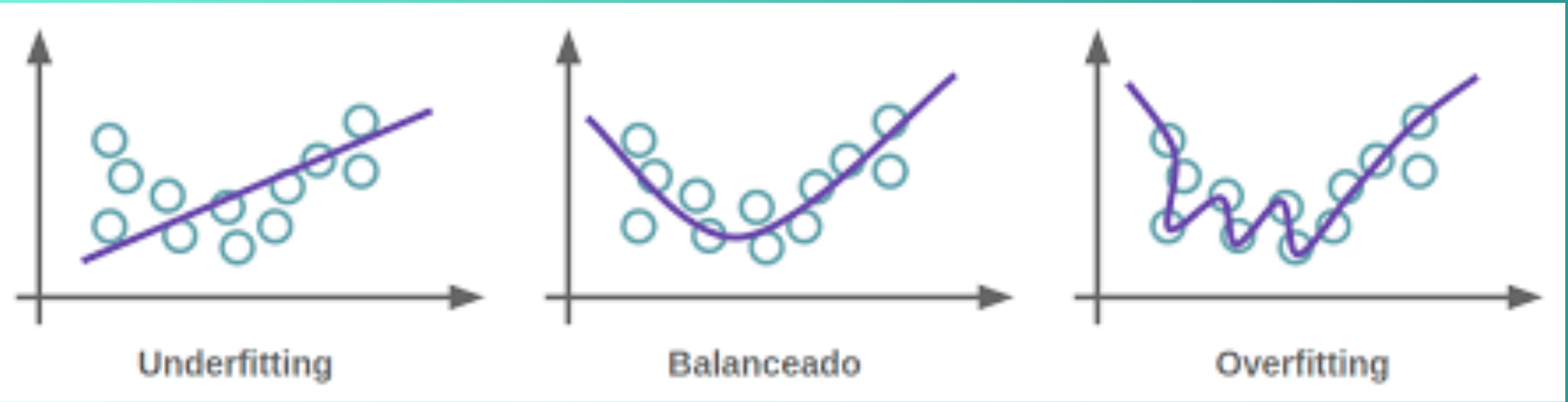
Autoencoders

- An autoencoder is an unsupervised artificial neural network that acquires the skill of effectively compressing and encoding data.
- It subsequently learns the process of reconstructing the data from the reduced encoded representation, aiming to create a representation that closely resembles the original input.
- It decreases data dimensions by acquiring the capability to filter out the noise present in the data.



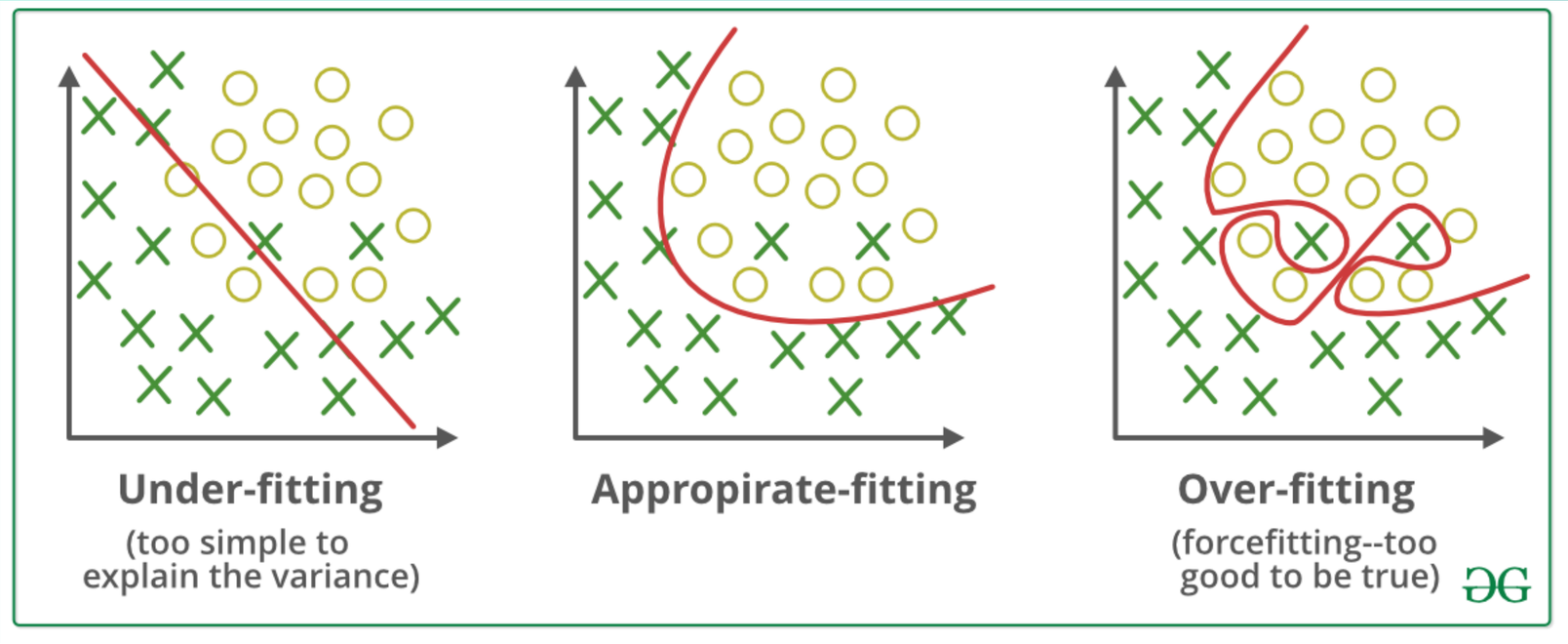
Overfitting

- A crucial aspect of training a machine learning model is **preventing overfitting**.
- **Overfitting**: Good prediction on training data and bad generalization to new data.
- **Underfitting**: Poor prediction on training data and bad generalization to new data.



Regularization

- Regularization is used to prevent overfitting and improve the generalization of models.



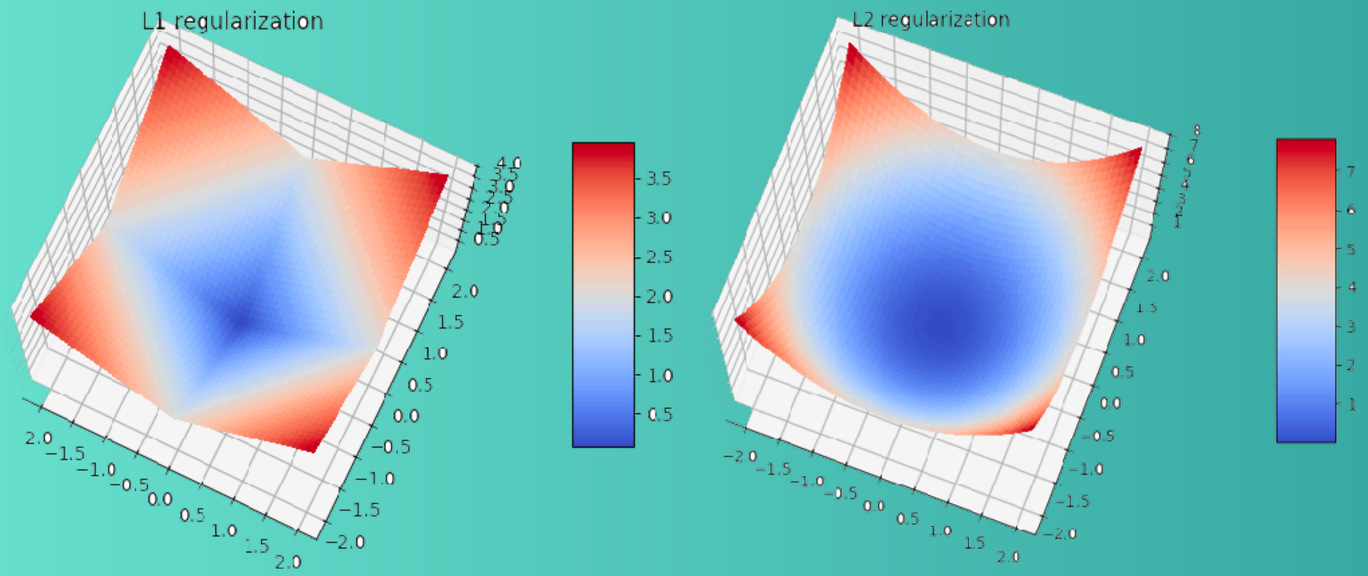
Regularization

- Regularization is effective in scenarios where we have numerous features that provide only marginal utility.
- Regularization adds a penalty term to the cost function to discourage complex models.
- Leveraging L1 and L2 regularization represents optimal methods to manage overfitting and conducting feature selection within an extensive feature set.
- The most common form of regularization is L2 regularization - Ridge, which adds the squared magnitude of the coefficient as a penalty term to the loss function.
- L1 regularization - Lasso is another form of regularization which can lead to sparse models where some feature weights are exactly zero.

Regularization

L1: Loss Function = Original Loss + $\lambda \sum |\beta_i|$

L2: Loss Function = Original Loss + $\lambda \sum \beta_i^2$



$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2$

Source: laid.delanover.com/

L1 Regularization - Lasso Regression

- Lasso - least absolute shrinkage and selection operator

$$\sum_{i=1}^{n}(Y_i - \sum_{j=1}^{p} X_{ij}\beta_j)^2 + \lambda \sum_{j=1}^{p} |\beta_j|$$

RSS - residual sum of squares

Regularization Term - Penalty

y_i : Actual value

$\hat{y}_i$: Predicted value

β_j : Model coefficients

λ : Regularization strength

L2 Regularization - Ridge Regression

- Ridge regression adds the “squared magnitude” of the coefficient as the penalty term to the cost function

$$\sum_{i=1}^{n}(Y_i - \sum_{j=1}^{p} X_{ij}\beta_j)^2 + \lambda \sum_{j=1}^{p} \beta_j^2$$

RSS - residual sum of squares

Regularization Term - Penalty

y_i : Actual value

$\hat{y}_i$: Predicted value

β_j : Model coefficients

λ : Regularization strength

Lasso Regression vs. Ridge Regression

- Key Observations
- The Trade-offs:
  - Interpretability vs. Performance
  - Feature Correlation
  - Effect of Regularization Strength (λ)
- Practical Applications

summary

- Applying L1 regularization to a regression model, can produce a model where many of the feature coefficients are zero (sparsity), thereby reducing the number of features used in the final model. This is a form of embedded feature selection, which can lead to simpler, more interpretable models that may generalize better to new data.
- L2 regularization is a technique used to improve the generalization of linear regression models by penalizing large coefficients, thus reducing overfitting. All features are kept in the model, but their coefficients are reduced in magnitude, resulting in a more stable and reliable model, especially when dealing with correlated predictors or when the number of predictors is large compared to the number of observations.